

Hardware and software

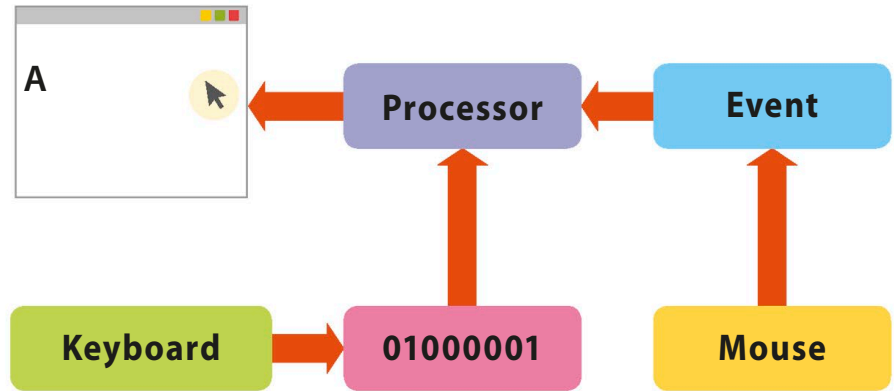
The physical parts of a computer that exist as objects in the real world are known as hardware. The combination of instructions and data telling hardware to perform tasks is called software. Software is also known as code or programs. Writing software can be challenging, as it involves writing instructions for a machine with no real understanding of the world.

▽ Mouse input

A computer classes every mouse movement as an “event”. Its operating system is constantly checking for events and reacts to each one by running code to deal with it specifically.

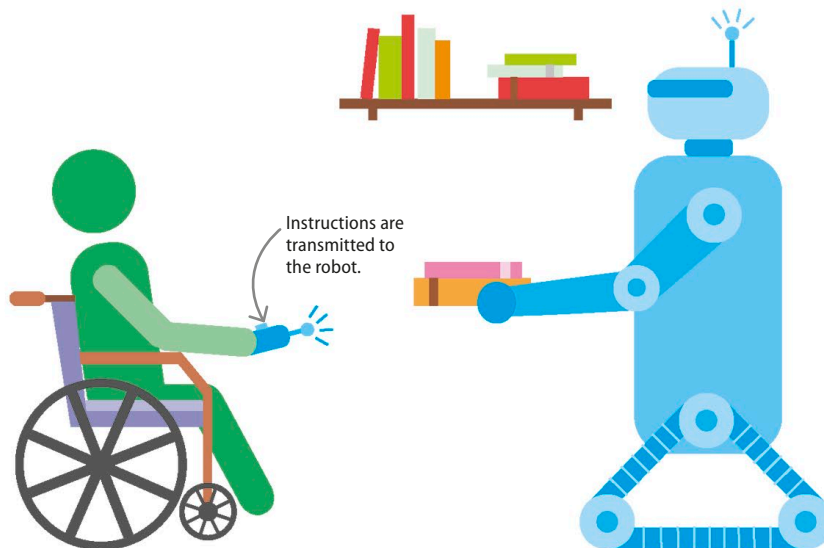
▷ Keyboard input

A user pressing a key on the keyboard is an event that computers recognize. Programmers often use these events to make something happen on the monitor, such as writing a letter.



Physical computing

Computers are increasingly affecting objects in the physical world, from robot helpers to apps that can control lights in people's homes. This is made possible by software containing instructions that turn numbers into signals. For instance, a smartphone app may send a signal to a nearby Wi-Fi router. This travels across the internet to trigger a light in the user's home.



REAL WORLD

How much code?

The size of programs is usually measured in lines of code. This is how many lines of code are in some well-known pieces of software:

Simple mobile app: 10,000

NASA's space shuttle: 400,000

Boeing 787 plane: 6.5 million

Firefox web browser: 9.7 million

Microsoft Office 2013: 45 million

Google: 2 billion

◁ Robots

A control device for a robot may send instructions as radio signals that, when received by the robot, are translated into electrical signals that operate the robot's motors and gears.